UNIVERSITY OF PIRAEUS

DEPARTMENT OF DIGITAL SYSTEMS

MSc Program: Information Systems and Services

Specialization: Big Data and Analytics

Course: Big Data Programming and Infrastructures: Python and Cloud Computing

Prediction of Drug-Induced Autoimmunity (DIA)

Student Team:

Marina Nouchou - Student ID: 2423

Panagiotis Athanasopoulos - Student ID: 2402

Academic Year: 2024-2025

Contents

1. Introduction

Predicting Drug-Induced Autoimmunity (DIA) is a critical challenge in drug toxicology, since rare but serious autoimmune reactions can lead to drug withdrawal or severe adverse effects. The dataset includes 597 drugs (148 positive and 449 negative) and constitutes a binary classification problem with substantial class imbalance (approximately 25% positive samples). Recent studies have reported strong performance on this dataset. The InterDIA platform proposed an integrated and interpretable DIA prediction framework based on molecular physicochemical properties. It used multi-strategy feature selection, advanced ensemble resampling, an Easy Ensemble Classifier, SHAP-based interpretation, and a Streamlit web platform. Another comparative study evaluated XGBoost, CatBoost and Gradient Boosting and highlighted the importance of metrics such as F1-score and MCC in addition to accuracy. The objective of this work is to optimize DIA prediction models by extending existing approaches. Through appropriate preprocessing, feature selection, imbalance-handling techniques and multiple classifiers (XGBoost, Random Forest and Logistic Regression), the study aims to achieve an F1-score above 75%, while emphasizing interpretability and prediction reliability through SHAP, confusion matrices and calibrated probabilities.

2. Dataset

The dataset used in this study originates from the UCI Machine Learning Repository and concerns the prediction of Drug-Induced Autoimmunity (DIA). It contains 597 drug molecules described by 195 continuous molecular descriptors extracted using RDKit. Each sample contains 195 numerical descriptors representing molecular, physicochemical, topological, electronic and structural properties; a SMILES string representing chemical structure; and a binary Label, where 1 denotes a DIA-positive molecule and 0 a DIA-negative molecule. The dataset is already divided into a training set of 477 samples (118 positive, 359 negative) and a test set of 120 samples (30 positive, 90 negative). The positive-to-negative ratio is approximately 1:3 in both sets. No missing values were identified.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the main characteristics of the DIA dataset, identify skewness and scale differences, assess the need for normalization, and prepare the data for modeling.

3.1 Dimensions and Basic Statistics

The original dataset contains 597 drug molecules, each represented by 195 RDKit molecular descriptors, together with SMILES and the binary target Label. The training set contains 477 records (118 positive, 359 negative), while the test set contains 120 records (30 positive, 90 negative). No missing or invalid values were found.

3.2 Class Distribution

The training data are clearly imbalanced: 359 DIA-negative samples (75%) and 118 DIA-positive samples (25%). This imbalance motivates the use of F1-score, Recall and ROC AUC rather than relying only on Accuracy.

3.3 Descriptive Statistics and Normalization

Descriptive statistics revealed substantial heterogeneity in descriptor scales and evidence of skewness and outlying values. The original plots and numerical results are preserved below exactly as in the source report.

3.4 Comparative Overview of Descriptors

Selected descriptors such as BalabanJ, BertzCT, Chi0 and Chi0n were compared to identify extreme values and potentially redundant information. BertzCT exhibits particularly high values, while Chi0 and Chi0n remain close to zero in the displayed statistics.

3.5 Principal Component Analysis (PCA)

PCA was applied for dimensionality reduction and visual inspection. Approximately 90% of the information is retained by the first ~75 components, while 95% is covered by fewer than 100 components. The projection onto the first two principal components does not show clear class separation, suggesting the need for nonlinear classifiers.

4. Methodology

The DIA prediction workflow included preprocessing, incorporation of chemical information through a Tanimoto similarity graph, imbalance handling, feature engineering, model training and final evaluation.

4.1 Preprocessing and Normalization

The descriptors were normalized and features with zero variance, duplicate columns and highly correlated features (correlation > 0.95) were removed to reduce dimensionality and redundancy.

4.2 Feature Enrichment: Tanimoto Graph Embeddings

Tanimoto-based similarity graphs were constructed from ECFP6 fingerprints. Laplacian Eigenmaps embeddings (dimension 64-96) were extracted, graph smoothing was applied to the descriptors, and the top ECFP bits selected using the chi-squared statistic were added to the final feature representation.

4.3 Class Imbalance Handling

• SMOTE oversampling of the minority class (ratio = 0.8).

• Scaffold-aware cross-validation based on Murcko scaffolds to reduce data leakage.

• No class weights when SMOTE is used, avoiding double weighting.

• Dual-threshold classification for uncertain predictions, with minimum coverage and F-beta optimization.

• Isotonic calibration of predicted probabilities to reduce overconfidence.

• Seed averaging using five repetitions per model with different random seeds.

4.4 Feature Engineering

• SHAP values based on XGBoost for feature contribution analysis.

• Recursive Feature Elimination (RFE) for progressive feature removal.

• Genetic selection using DEAP with evolutionary optimization for F1-score.

4.5 Models and Training

Three main models were trained: Logistic Regression with elastic-net regularization, Random Forest with 1,000+ trees and optimized hyperparameters, and GPU-accelerated XGBoost tuned with RandomizedSearchCV. All models were evaluated with 5-fold scaffold cross-validation and calibrated probabilities.

5. Results

5.1 Comparative Metrics Table

Model	Version	Accuracy	Precision	Recall	F1-score	ROC AUC
XGBoost	Seed-Avg	0.8917	0.8696	0.6667	0.7547	0.9274
XGBoost	Single-Fit	0.8750	0.8261	0.6333	0.7170	0.9044
Random Forest	Seed-Avg	0.8917	0.8148	0.7333	0.7719	0.9326
Random Forest	Single-Fit	0.8750	0.8000	0.6667	0.7273	0.9357
Logistic Regression	Seed-Avg	0.8583	0.8824	0.5000	0.6383	0.8770
Logistic Regression	Single-Fit	0.8583	0.8824	0.5000	0.6383	0.8770

Table 1: Comparative metrics for XGBoost, Random Forest and Logistic Regression (Test Set). All numerical results are unchanged from the original report.

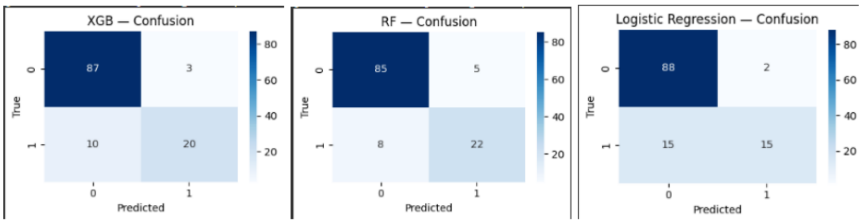
5.2 Results Analysis

XGBoost achieved high Precision (0.8696) and ROC AUC (0.9274), but lower Recall (0.6667). Random Forest provided the best balance between Precision and Recall, with Recall = 0.7333, F1-score = 0.7719 and ROC AUC = 0.9326. Logistic Regression achieved the highest Precision (0.8824) but substantially lower Recall (0.5000), making it a useful interpretable baseline but less sensitive to DIA-positive cases.

5.3 Confusion Matrices

Drug-Induced Autoimmunity ΜΔΑ 2024-2025

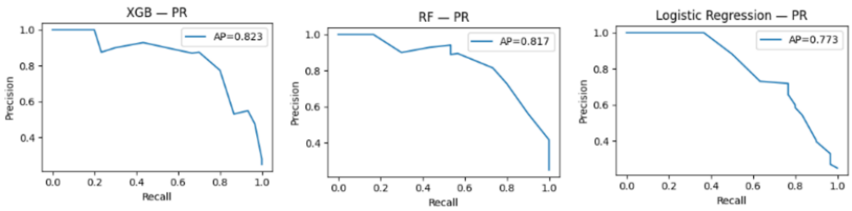
5.3. Confusion Matrices



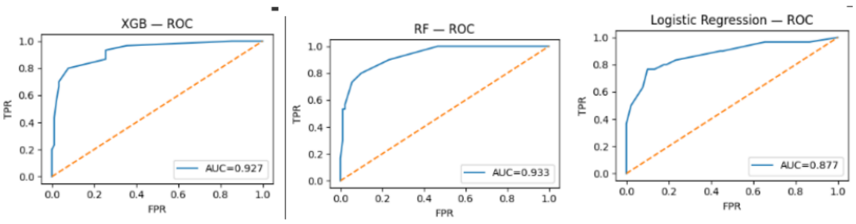
Σχήμα 7: Confusion Matrices για XGBoost, Random Forest και Logistic Regression αντίστοιχα

- Το XGBoost είχε 10 False Negatives (FN), μειώνοντας το recall.
- Το Random Forest είχε μόνο 8 FN και ισορροπημένα True Positives (TP).
- Η Logistic Regression είχε 15 FN και 15 TP, ενδεικτικό χαμηλής ευαισθησίας.

5.4. Καμπύλες Precision-Recall και ROC



Σχήμα 8: Καμπύλες Precision-Recall για XGBoost (AP=0.823), RF (AP=0.817) και LR (AP=0.773)



Σχήμα 9: ROC καμπύλες για XGBoost (AUC=0.927), RF (AUC=0.933) και LR (AUC=0.877)

Η καμπύλη PR του **XGBoost** δείχνει υψηλό precision σε όλο το φάσμα recall, ενώ η **Random Forest** διατηρεί καλύτερη ισορροπία. Η **Logistic Regression** παρουσιάζει απότομη πτώση του precision με αύξηση του recall, δείχνοντας χαμηλότερη αξιοπιστία σε περιοχές υψηλής κάλυψης.

Η καμπύλη ROC του **Random Forest** είναι πιο «τετραγωνισμένη», κοντά στο ιδανικό σημείο (0,1), επιβεβαιώνοντας ότι πρόκειται για το πιο ισχυρό μοντέλο στο συγκεκριμένο dataset.

9

Original confusion matrices, Precision-Recall curves and ROC curves (source report, page 10). The plots and reported values are preserved unchanged.

The XGBoost confusion matrix contains 10 false negatives, Random Forest contains 8 false negatives, and Logistic Regression contains 15 false negatives. The corresponding PR average-precision values are 0.823, 0.817 and 0.773, while ROC AUC values are 0.927, 0.933 and 0.877, respectively.

6. Conclusions

This study developed and evaluated classification models for identifying DIA-positive cases in an imbalanced molecular dataset. Particular emphasis was placed on class imbalance, prediction reliability and comparison across algorithms. Random Forest achieved the best overall balance, with F1-score = 0.7719 and ROC AUC = 0.933. XGBoost achieved high Precision = 0.8696 but lower Recall = 0.6667. Logistic Regression, while simpler and more interpretable, produced Recall = 0.5000 but high Precision = 0.8824 and AUC = 0.877. Seed averaging helped stabilize results, isotonic calibration improved the reliability of probability estimates, and the dual-threshold strategy introduced a low-confidence region for uncertain predictions. For applications where detecting DIA-positive compounds is critical, Random Forest is the most suitable of the evaluated models; where minimizing false positives is more important, XGBoost may be preferable.

References

[1] M. Belkin and P. Niyogi, Laplacian Eigenmaps for Dimensionality Reduction and Data Representation, Neural Computation, 15(6), 1373-1396, 2003.

[2] M. Buda, A. Maki, and M. A. Mazurowski, A Systematic Study of the Class Imbalance Problem in Convolutional Neural Networks, Neural Networks, 106, 249-259, 2018.

[3] N. V. Chawla et al., SMOTE: Synthetic Minority Over-sampling Technique, Journal of Artificial Intelligence Research, 16, 321-357, 2002.

[4] Delfiero et al., Comparative study of ML algorithms for predicting drug-induced autoimmunity using molecular descriptors, Jurnal Mandiri IT, 14(1), 180-190, 2025.

[5] T. G. Dietterich, Ensemble Methods in Machine Learning, Multiple Classifier Systems, Springer, 1-15, 2000.

[6] C. Elkan, The Foundations of Cost-Sensitive Learning, IJCAI, 2001.

[7] G. Forman, An Extensive Empirical Study of Feature Selection Metrics for Text Classification, JMLR, 3, 1289-1305, 2003.

[8] F.-A. Fortin et al., DEAP: Evolutionary Algorithms Made Easy, JMLR, 13, 2171-2175, 2012.

[9] I. Guyon et al., Gene Selection for Cancer Classification using Support Vector Machines, Machine Learning, 46, 389-422, 2002.

[10] Huang et al., InterDIA: Interpretable Prediction of Drug-induced Autoimmunity through Ensemble Machine Learning Approaches, Toxicology, 511, 2025.

[11] S. M. Lundberg and S.-I. Lee, A Unified Approach to Interpreting Model Predictions, NeurIPS 30, 4765-4774, 2017.

[12] P. G. Polishchuk et al., Estimation of the Size of Drug-like Chemical Space Based on GDB-17 Data, Journal of Computer-Aided Molecular Design, 27(8), 675-679, 2013.

[13] D. Rogers and M. Hahn, Extended-Connectivity Fingerprints, Journal of Chemical Information and Modeling, 50(5), 742-754, 2010.

[14] UCI Machine Learning Repository. Drug Induced Autoimmunity Prediction Dataset.

[15] B. Zadrozny and C. Elkan, Transforming Classifier Scores into Accurate Multiclass Probability Estimates, ACM SIGKDD, 694-699, 2002.

[16] D. Zhou et al., Learning with Local and Global Consistency, NeurIPS 16, 321-328, 2004.